

GE Healthcare

LightSpeed 1.X-4.X

Quick Tube Change Procedure

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The information in this manual applies to all GE Healthcare LightSpeed 1.X through 4.X CT Scanners.

IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

(EN)

- This Service Manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.

Предупреждение

(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告

(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

- Tento provozní návod existuje pouze v anglickém jazyce.
- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL

(DA)

- Denne servicemanual findes kun på engelsk.
- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING

(NL)

- Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

HOIATUS

(ET)

- Käesolev teenindusjuhend on saadaval ainult inglise keeles.
- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS

(FI)

- Tämä huolto-ohje on saatavilla vain englanniksi.
- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION

(FR)

- Ce manuel de service n'est disponible qu'en anglais.
- Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

(DE)

- Diese Serviceanleitung existiert nur in Englischer Sprache.
- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, Mechanische oder Sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

- Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο.
- Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

(HU)

- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.
- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

(IS)

- Þessi þjónustuhandbók er eingöngu fánleg á ensku.
- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA

(IT)

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告

(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다 .
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우 , 번역 서비스를 제공하는 것은 고객의 책임입니다 .
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 .
- 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .

BRDINJUMS

(LV)

- Šī apkalpes rokasgrāmata ir pieejama tikai anglu valodā.
- Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis anglu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS

(LT)

- Šis eksploatavimo vadovas yra prieinamas tik anglų kalba.
- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė.
- Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote.
- Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.

ADVARSEL

(NO)

- Denne servicehåndboken finnes bare på engelsk.
- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE

(PL)

- Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.
- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.

ATENÇÃO

(PT)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente consertar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode pôr em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.

ATENȚIE

(RO)

- Acest manual de service este disponibil numai în limba engleză.
- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!

(RU)

- Данное руководство по обслуживанию предлагается только на английском языке.
- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.

UPOZORNENIE

(SK)

- Tento návod na obsluhu je k dispozícii len v angličtine.
- Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.

ATENCION

(ES)

- Este manual de servicio sólo existe en inglés.
- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

- Den här servicehandboken finns bara tillgänglig på engelska.
- Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DIKKAT

(TR)

- Bu servis kılavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://egems.med.ge.com/edq/home.jsp>
- Contact your local service coordinator for more information on this process.

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CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION



Risk of Explosion.

Danger of explosion if battery is incorrectly replaced.

Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.

ATTENTION



Danger d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie.

Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the Healthcare PQR Process to report all omissions, errors, and defects in this publication.

TRADEMARKS

All products and their name brands are trademarks of their respective holders.

Revision History

Revision	Date	Reason for change
0	09/23/03	Initial Release
1	01/12/04	Chapter 2: Added Tube Bolt Reuse Avoidance Statement. Chapter 3: Updates to Tube & Mounting Plate torque requirements. Chapter 4: Added Section 13.0 - “Gantry Rotation Safety Check”
2	05/13/04	General Updates.
3	04/24/06	Update for Tube Installation Certification.
4	12/7/07	PQR 13131422: Updated Chapter 4, Section 3.5.8: "Install New Tube Program," step 6; removed Table 4-2. Important Precautions: added new language warnings

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Chapter 1

Introduction

Section 1.0

Purpose

This document covers the quick replacement procedure for X-Ray Tubes. See [Figure 1-1](#) for an overview of this process. The sections following explain this process.

Section 2.0

Requirements

The system must meet the following requirements to perform the Quick Tube Replacement Procedure:

- 1.) Normal Tube Change with no other system issues.
- 2.) The previous Tube was aligned correctly. (POR, BOW, CBF, ISO were in spec.)
- 3.) The Collimator and the Detector have not moved.
- 4.) There are no IQ issues.
- 5.) If kV gain pots are adjusted, you must run a full set of calibrations.

If these requirements are met, the Quick Tube Replacement Procedure can be followed. If the system does not meet these requirements, the full Tube Change Procedure described in the LightSpeed 4.X System Service Manual must be performed.

Tube Installation Certification must be performed after changing the tube and after performing BOW, Hot ISO Alignment or Cold ISO Alignment. If the Certification procedure is not performed and the installed tube is a GE Performix tube, then GE Performix Tube "Advisory" messages will be posted through pop-ups and in the Dose Report screen.

Section 3.0

Quick Tube Replacement Process Flow

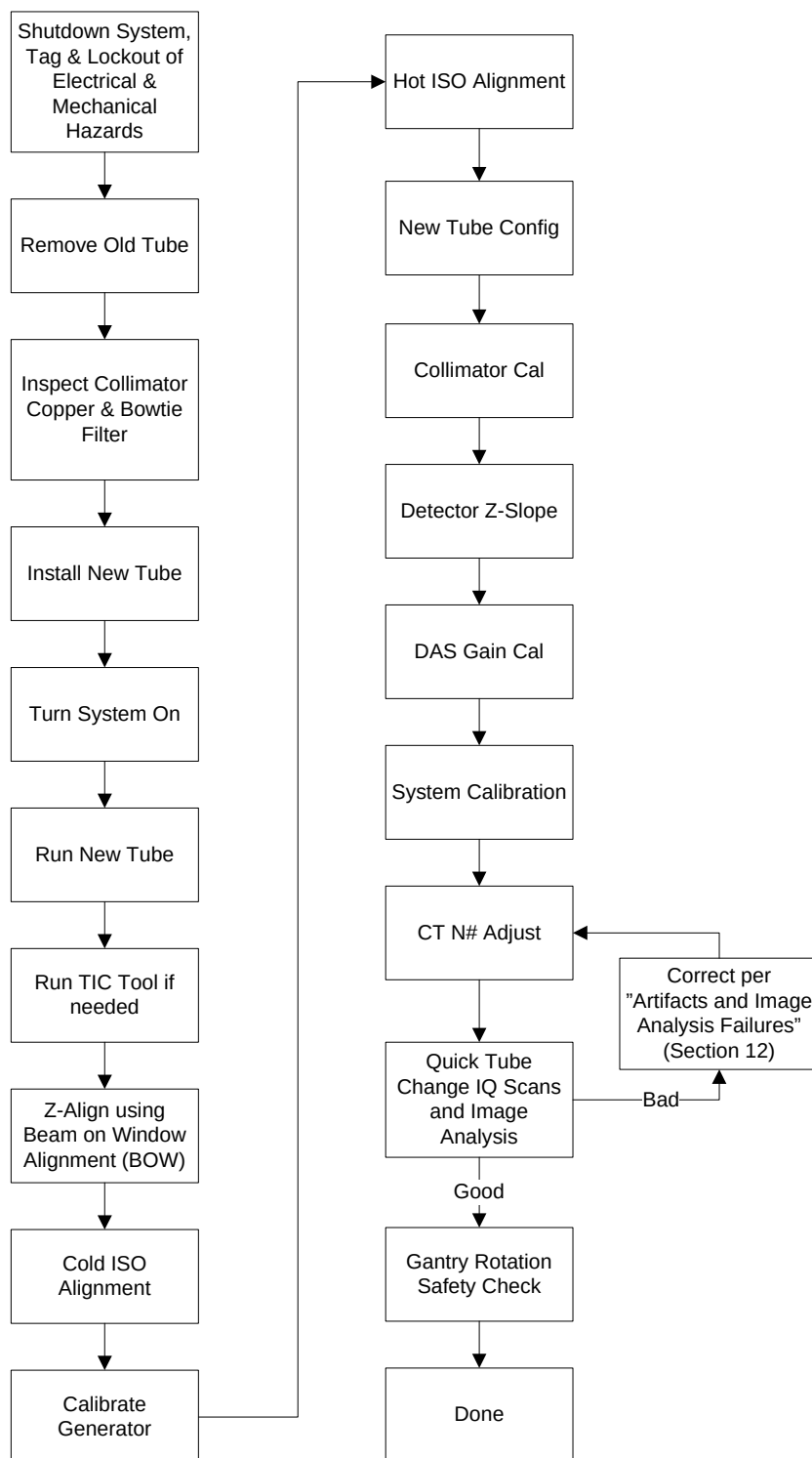


Figure 1-1 Quick Tube Change Flow

Chapter 2

Remove Old Tube

Before beginning this procedure, please read the safety information.

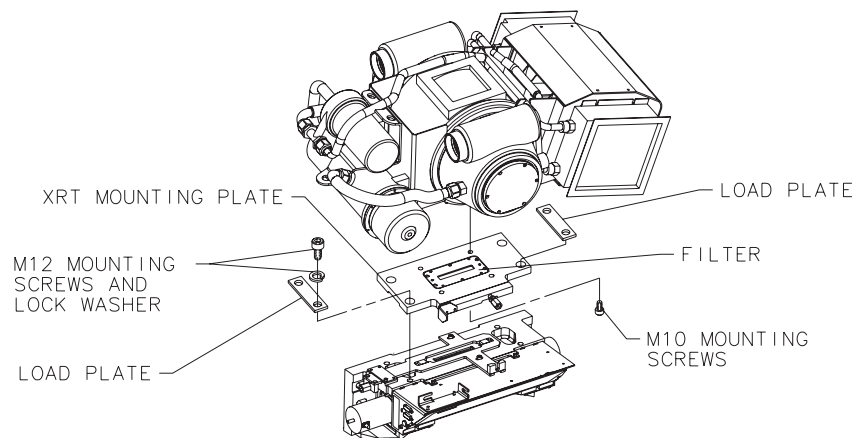


Figure 2-1 Tube Removal Diagram



WARNING



FOLLOW LOCKOUT/TAGOUT PROCEDURES FOR POWER & ROTATIONAL HAZARDS.

Make sure you engage the locking mechanism before you remove the tube. Failure to lock the gantry could result in injury, should the gantry suddenly move and strike you.

- 1.) Remove, and set aside, both gantry side covers, top covers, front and rear covers.
- 2.) Remove the M12 screws from the right front gantry cover mounting bracket, remove and set aside the bracket. Reference [Figure 2-2](#). (Bracket is not part of the LightSpeed 1.X Gantry.)



NOTICE

It might be necessary to tilt the gantry back to remove the third bolt, which is not normally installed. Remember to tilt the gantry back to zero degrees.



Figure 2-2 Right Front Gantry Cover Mounting Bracket (not installed on LS 1.X)

- 3.) Turn OFF facility power to **PDU** and lockout/tagout.
- 4.) Turn OFF the **AXIAL DRIVE ENABLE** and **HVDC ENABLE** or **550 VDC ENABLE** service switches.



NOTICE

It may be easier to loosen the M12 tube mounting bolts with the tube at about the 2 o'clock position before locking the tube at the 3 o'clock position. Simply loosen the tube mounting bolts one half (1/2) turn. Do not remove the bolts yet.

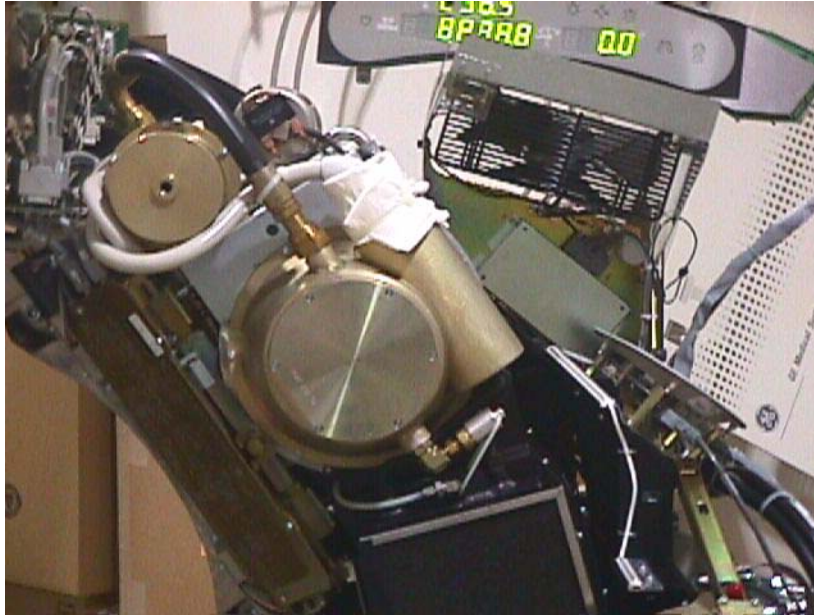


Figure 2-3 Tube Angle to Loosen and Torque M12 Screws (LS 2.X-4.X shown)

- 5.) Rotate the Gantry until the failed tube unit reaches the 3 o'clock position.



NOTICE

Make sure the tube is at 90 degrees so the tube hangs at the correct engagement angle for removal and installation.

- 6.) Engage rotational lock. Check that the gantry is securely locked by attempting to rotate the gantry by hand.
- 7.) **All systems EXCEPT LightSpeed 1.X:**
Insert the lifting post, boom and chain hoist. Reference [Figure 2-2](#).
LightSpeed 1.X systems:
Set up the chain hoist
- 8.) Disconnect the 12 pin tube I.D. system cable, from the top of the tube unit.
- 9.) Disconnect the 4 pin mate-n-lock pump and fan power system cable
- 10.) Disconnect the ground strap from the top of the tube unit
- 11.) Remove the anode and the cathode cable:
 - Carefully cut ty-raps securing **HV** cables. Note **HV** cable routing.
 - Loosen each cable's locking ring with the spanner wrench.
 - Pull each cable terminal out of its receptacle.
 - Ground the end of the cables to the Gantry frame.
 - Wipe up any oil that drips from the cable terminal.
 - Use paper towels to soak up any oil in the wells.



CAUTION



Remove the mounting bars in the following (lower/upper) order to lessen the risk of injury to your hand. Keep one hand under the bolt and pressure plate while unfastening it. This is to prevent them from falling into the fan that is attached to the tube.

Note: It may be easier to tape the socket to the extension. This will prevent the socket from being dislodged on the tube radiator assembly

12.) The XRT is attached to the Collimator with a Tube Mount Bracket Assembly (P/N 2128696). Remove the mounting plate & XRT from the Collimator by removing the four M12 (P/N 46-328416P24) cap screws and lock washers, and two load plates (P/N 2120104), with a hex socket driver. With your hand, reach behind the radiator to the screws from either side of the XRT center section while removing the bolts with two 12 inch extensions (24 inch length) on a ratchet. **Throw these M12 bolts and washers away, as they must not be reused.**

13.) Carefully swing the tube clear of the gantry.

Note: Be careful not to damage the fragile copper filter or lead shield in the mounting plate for the next step.



WARNING



SEVERE INJURY POSSIBLE.

TUBE CAN SEPARATE FROM GANTRY

DO NOT REMOVE FACTORY INSTALLED MOUNTING PLATE.

14.) If the replacement tube has a mounting plate attached (see [Figure 2-4](#)), DO NOT REMOVE IT. Skip steps 15 and 16.

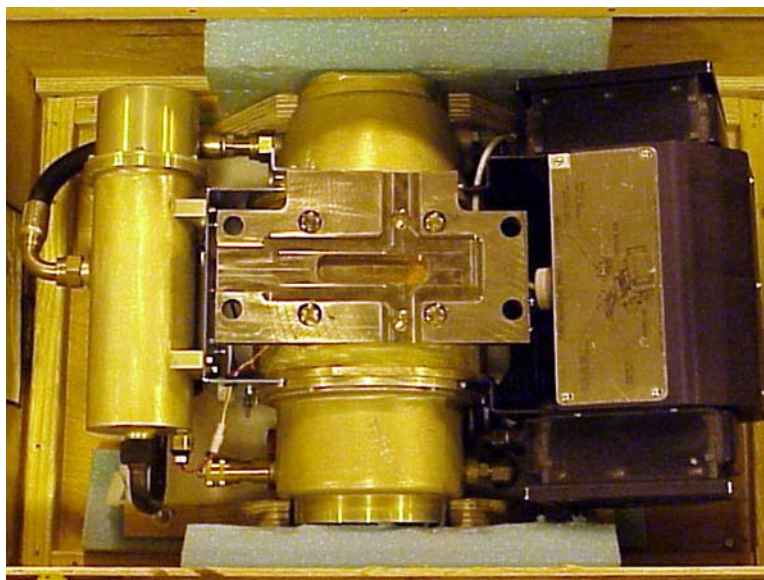


Figure 2-4 Tube Shipped with Mounting Plate Attached



NOTICE
Potential for IQ artifacts

When removing the mounting plate from the tube, be careful with the Copper Filter. It should be free of debris, scratches and dust. Particles create artifacts in the image by affecting the attenuation properties of the copper filter.

15.) If the replacement tube does NOT have a mounting plate attached, remove the mounting plate by removing the four M10 (P/N 46-328416P20) hex head screws. **Throw these bolts and washers away, as they must not be reused.**

16.) If new tube has no mounting plate attached, inspect the copper filter. The copper filter should be clean, dent and scratch free:

- If contamination is visible (see [Figure 2-5](#)), proceed to the cleaning process (refer to the “Artifacts

Caused by Collimator Grease” in the appropriate *System Service Manual*).

- Discoloration is acceptable.



Figure 2-5 Extremely Contaminated Copper Filter

- 17.) Inspect the Bowtie Filter. It is possible that the Bowtie Filter is contaminated and the Primary Copper Filter is not contaminated.

Note: Perform this inspection before installing the new tube unit. Also look at the tube side of the copper filter when you are swapping the interposer plate.

The following tools are required for this inspection procedure:

- Phillips #2 Screwdriver
- Flat blade Screwdriver
- Bright Flashlight

Inspect the Bowtie Filter as follows:

- a.) Remove the Collimator Control Board Sheet Metal Cover. See [Figure 2-6](#).



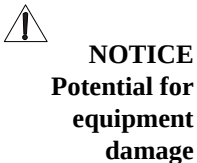
Figure 2-6 Collimator Control Board Cover



WARNING

DO NOT ATTEMPT TO MOVE THE FILTER WITH POWER ON.

- b.) With power removed from the gantry, use a flat blade screwdriver and position the bowtie filter assembly so it is visible through the input port or tube side of the collimator assembly.
- * CCW will move the filter into the beam. See [Figure 2-7](#).
 - * Do not move the filter back to the home position.



NOTICE
Potential for
equipment
damage

Do not force the filter if it feels stuck. Damage to the limit switch can result.

Do not move the filter more than necessary for inspection. The filter can fall off the drive screw.

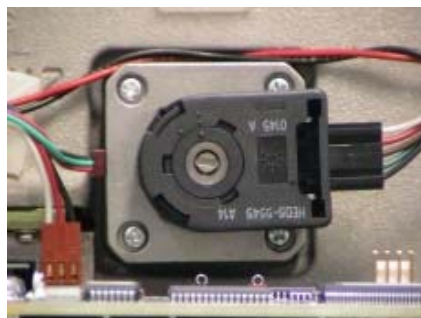


Figure 2-7 Filter Position Adjuster

- c.) Using a flashlight, inspect the bowtie filter for contamination.
Look through the input port or tube side of the collimator.
If contamination is visible, proceed to cleaning process.

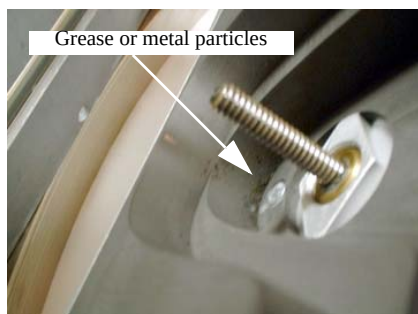


Figure 2-8 Contaminated Bowtie Filter

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Chapter 3

Install New Tube



WARNING

SEVERE INJURY POSSIBLE.

TUBE CAN SEPARATE FROM GANTRY, IF NOT TORQUED CORRECTLY.

ALWAYS USE PROPER TORQUE ON ALL FASTENERS.



WARNING

FOLLOW LOCKOUT/TAGOUT PROCEDURES FOR POWER & ROTATIONAL HAZARDS.



- 1.) Allow the tube unit to warm to room temperature before you install it.



WARNING

SEVERE INJURY POSSIBLE.

TUBE CAN SEPARATE FROM GANTRY

DO NOT REMOVE FACTORY INSTALLED MOUNTING PLATE.



- 2.) If the new tube has a factory installed mounting plate, DO NOT REMOVE IT. Skip to Step 4.
- 3.) Attach the mounting plate from the old tube using **four new M10 bolts** that come with the tube. **Do NOT use Loctite.**
 - Finger tighten all four (4) bolts
 - Set all four (4) bolts to the following pre-load torque specification:

15 lb-ft.	20 N-m	180 in-lbs	210 kg-cm
-----------	--------	------------	-----------

This seats each bolt, enabling you to visually ensure that the mounting holes are not stripped while applying final torque.



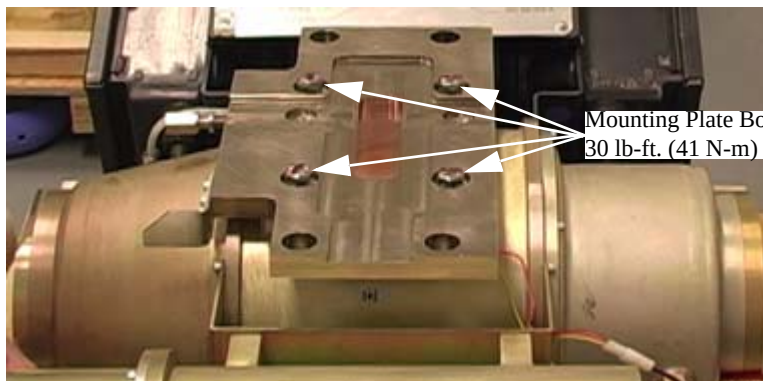
WARNING

SEVERE INJURY POSSIBLE.

IF YOU TURN THE TORQUE WRENCH MORE THAN 90° (¼ TURN) WHILE APPLYING FINAL TORQUE, THE MOUNTING THREADS ARE STRIPPED. DO NOT USE THIS TUBE.

- Set final torque on all four bolts:

30 lb-ft.	41 N-m	360 in-lbs	420 kg-cm
-----------	--------	------------	-----------



Mounting Plate Bolts - M10
30 lb-ft. (41 N-m) final torque

Figure 1 - Tube and Mounting Plate

CAUTION
Potential for IQ artifacts

When attaching the mounting plate on the tube, be careful with the Copper Filter. It should be free of debris, scratches and dust. The reason is particles create artifacts in the image by affecting the attenuation properties of the copper filter.

- 4.) Re-check facility power and make sure it is OFF.

WARNING

SEVERE INJURY TO PATIENT COULD OCCUR.
USING THE WRONG BOLTS COULD PLACE STRESS ON BOLTS, CAUSING TUBE TO SEPARATE FROM GANTRY.
USE PROPER BOLTS, PER SYSTEM TYPE.

- 5.) Use the hoist to lift the new tube unit:
- a.) Position the tube on the gantry: The “crosses” on the mounting plate and on the collimator should fit in perfectly when the tube is aligned properly.

Note:

- * To ease installation, fasten the top pressure plate to the rotating structure first. Then attach the bottom pressure plate.
- * **Use new bolts and washers from tube crate. Make sure to select the proper bolts. There are instructions in the crate, and on the tube itself.**

- b.) Fasten the lower and upper pressure plates to the rotating structure with the four M12 (50 mm) bolts, and set **pre-load torque to:**

25 lb-ft.	33 N-m	295 in-lbs	340 kg-cm
------------------	---------------	-------------------	------------------

- c.) Set final torque to:

49 lb-ft.	66 N-m	590 in-lbs	680 kg-cm
------------------	---------------	-------------------	------------------

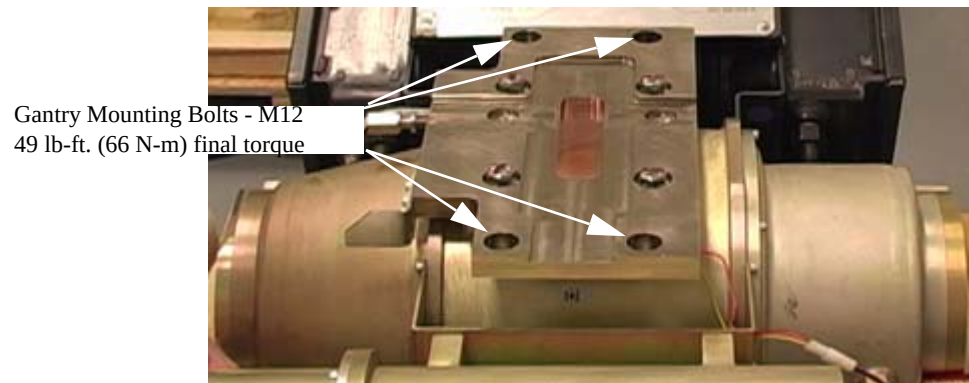


Figure 2 - Tube and Mounting Plate

- 6.) Attach the tube I.D. cable to the 12 pin mate-N-Lock connector on top of the tube.
- 7.) Attach the tube pump and fan power cable to the 4 pin mate-N-Lock connector.
- 8.) Fasten the grounding strap to the 1/4-20 ground stud on top of the tube unit.
- 9.) Remove the plastic cap plug from each cable receptacle on the tube unit.
- 10.) Lightly wet the new rubber quad ring with transformer oil (917).
- 11.) Return the quad ring to its slot at the top of the receptacle retaining ring.
- 12.) Pour transformer oil (917) into the receptacle to a depth of 10 mm (0.375 in).

Note:

Take care not to lose the rubber quad rings for the High Voltage cables.



NOTICE

Incorrectly routed or secured HV cables will result in damage to the HV cables and/or other parts of the gantry.

- 13.) Be sure to route the **HV** cable as shown in [Figure 3-3](#).

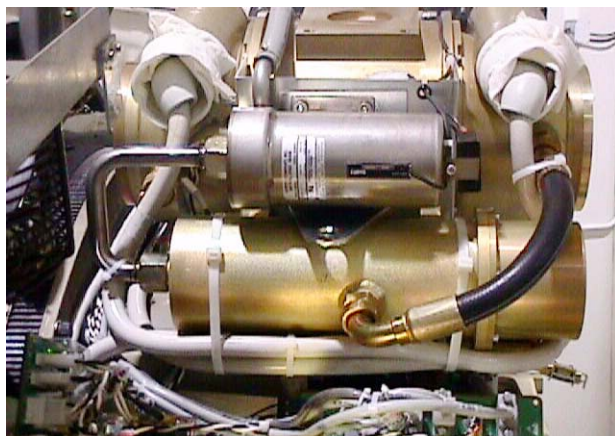


Figure 3-3 HV Cables Properly Routed and Secured

- 14.) Align the cable terminal orienting key with the notch in the receptacle.
15.) Slowly insert the cable, to engage the connector pins, and seat the cable in the well.
- Tighten the cable locking ring.
 - Rotate the cable strain relief for a clean cable dress.



NOTICE

Do not over tighten the locking ring. Over tightening can deform the cable plug sealing surfaces, break the oil seal between receptacle and housing, twist the receptacle, and disrupt internal wiring.



NOTICE

IF YOU GET OIL ON YOUR HANDS, WASH THEM NOW.

- 16.) Carefully wipe up all excess oil.
17.) Secure **HV** Cables using Large Ty-Raps, as shown in [Figure 3-3](#).
18.) Disconnect hoist from tube and boom.
19.) Remove the post and boom from the gantry. Reference [Figure 2-2, on page 19](#).
20.) Check for oil leaks:
- Wrap rags or paper towels around the cable horns, and tape them into place.
 - Manually rotate the tube to the 6 o'clock position.
 - Return the tube to the 3 o'clock position
 - Remove the toweling and wipe up all excess oil.
 - Wipe off the cable horns, locking rings, and strain reliefs with a rag dampened with alcohol.
 - Repeat with a dry rag.
 - Wrap the cable strain reliefs and locking rings with a single layer of absorbent paper tissue.
 - You can use two inch wide strips cut from a paper napkin.
 - Wrap the bottom edge of the paper around the top end of the cable horn, and tape it into place.
 - Extend the top edge of the paper over the top of the locking ring, and tape it to the plastic cable strain relief.
 - Remove paper after leak check.
- 21.) If oils leaks are found, tighten the locking ring slowly until there is no leakage, paying attention to not

over tighten.

- 22.) Install the right gantry front cover bracket. Reference [Figure 3-3, on page 27](#).
- 23.) Restore system power at the main disconnect panel.
- 24.) Turn on gantry 120 VAC, HVDC POWER or 550 VDC ENABLE, and AXIAL DRIVE ENABLE service switches. Wait at least 10 minutes to warm up the filament.

Chapter 4

Alignment and Calibration

Section 1.0

Z-Align using Beam on Window Alignment (BOW)

The purpose of the Z-Align procedure is to move the X-Ray Tube to be in alignment with the Detector and Collimator. The Detector and Collimator were aligned with the previous Tube, and Z-Align will correctly place the new Tube in the Z-Direction.

Note: For the BOW tool to work correctly, the X-Ray Tube needs to be close to proper alignment with the Collimator and Detector. If the Tube is not close to where it needs to be, the BOW tool will give erroneous data back. A POR scan and corresponding Tube movement can correct this problem.

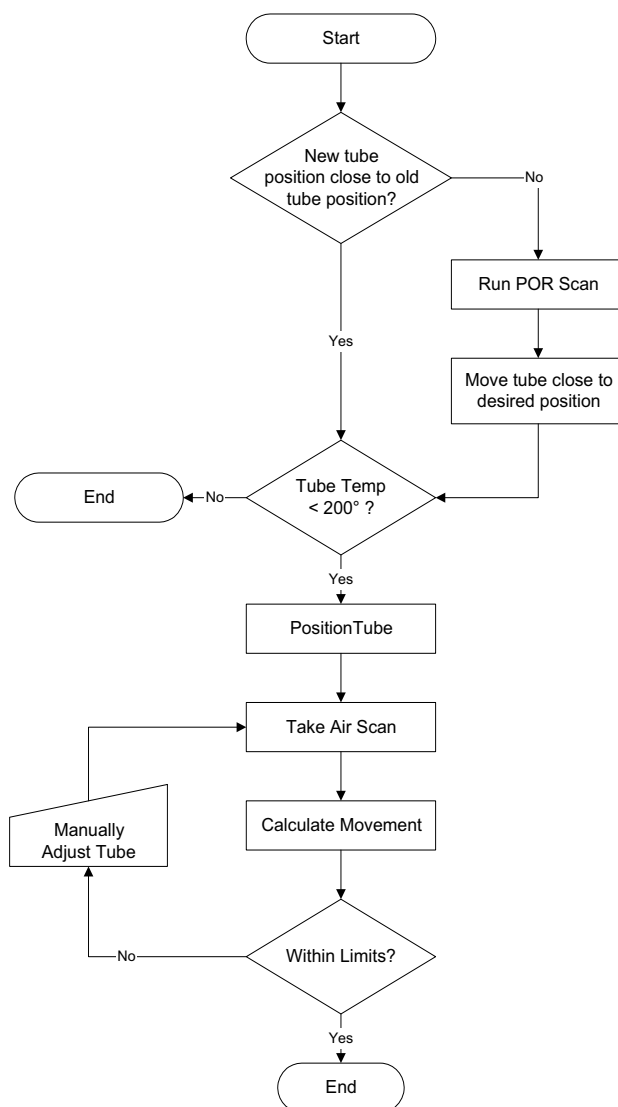


Figure 4-1 BOW Alignment Process Flow

1.1 Verify Tube Temperature < 200° C

- 1.) Click DAILY PREP.
- 2.) Click TUBE WARM-UP.
- 3.) Click ACCEPT in pop-up window.
- 4.) Click PAUSE.
- 5.) Click CANCEL (do not QUIT Daily Prep).



NOTICE Do not start X-ray exposures. Starting tube warm-up will result in additional tube cooling wait times.

- 6.) Open system error log (`gesys_suite.log`).
- 7.) Click LAST PAGE.

Look for this type of entry:

This entry is created when Pause is clicked.

```
bay57 dailyPrepRx
```

```
StateMachineEventNotify.c 418
```

Tube temperature before Cold Tube Warmup is 315.15 degrees Celsius.

This entry is created when Cancel is clicked.

```
bay57 dailyPrepRx
```

```
StateMachineEventNotify.c 411
```

Tube temperature after Cold Tube Warmup is 312.89 degrees Celsius.

Start Tube Alignments when 200.00 degrees Celsius is reported.

```
bay57 dailyPrepRx
```

```
StateMachineEventNotify.c 418
```

Tube temperature before Cold Tube Warmup is 200.00 degrees Celsius.

- 8.) Repeat steps 3 through 5 as necessary to generate updates in the `gesyslog`.
- 9.) When tube temperature shows 200.00 Celsius, click QUIT to exit Daily Prep.

1.2 Accessing the Software

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT
- 3.) Select TUBE CHANGE
- 4.) Select BOW ALIGNMENT

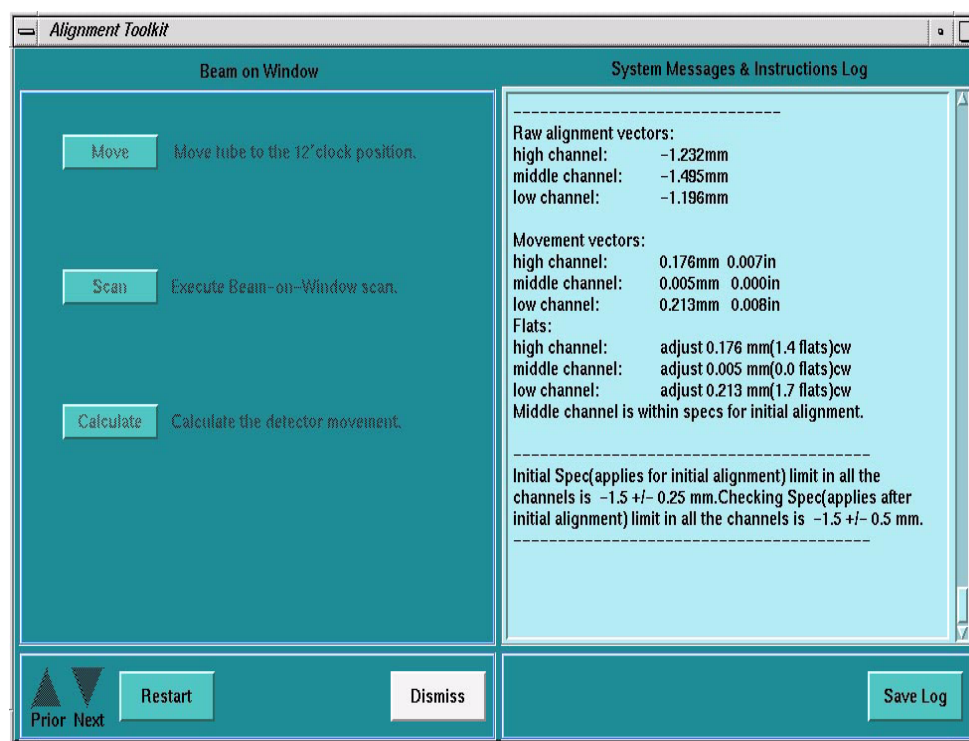


Figure 4-2 Screen Shot for Beam on Window Alignment

1.3 Adjustment Procedure

- 1.) Place a dial gauge in the POR Dial Mount location.
- 2.) Select MOVE to send the tube to the 12 o'clock position.
- 3.) Acquire a Beam on Window Scan.
- 4.) Select CALCULATE.
- 5.) If adjustments are required, calculate the Tube Z-Movement to bring BOW into spec.

BOW Movement vectors:

 - BOW high channel: -0.446mm -0.018in
 - **middle channel: -0.653mm -0.026in**
 - low channel: -0.611mm -0.024in
- 6.) After Selecting the CALCULATE button from the BOW Window, the GUI will display the BOW Raw Alignment Vectors, BOW Movement Vectors and the Flats Movement Vector.
- 7.) Tube Z-Movement Value = BOW Middle Channel Movement Vector divided by five.
- 8.) Tube Z-Movement Direction:
 - a.) If the BOW Movement Channel value is Positive (+) move the Tube towards the REAR of the Gantry. Turn the POR nut CCW.
 - b.) If the BOW Movement Channel value is Negative (-) move the Tube towards the TABLE. Turn the POR nut CW.

Example:

From the sample data above, the BOW Movement Vector for the Middle Channel is **-0.653mm or -0.026in.**

- Tube Z-Movement Value (mm) = -0.653 mm / 5 = -0.131 mm OR
- Tube Z-Movement Value (in) = -0.026 in / 5 = -0.0052 in.
- Tube Z-Movement Dir. negative = Towards the Table, turn the POR nut CW.

- 9.) Loosen the Tube Mounting Bolts.
- 10.) Move the Tube the desired Direction and Distance. Be as accurate as possible to avoid rescans.
- 11.) Tighten the Tube Mounting Bolts (49 ft lb)
- 12.) Select MOVE to send the tube to the 12 o'clock position.
- 13.) Acquire a Beam on Window Scan.
- 14.) Select CALCULATE
- 15.) If the Values are within spec, move on. If not, return to step 7.

Note: The specs for **BOW** are checked by the software. If an adjustment is needed for the first **BOW** scan, make adjustments per procedure. In the verification scan, after ISO alignments, the spec is different than the software version because the tube is warm. The spec is $-1.5 \pm 0.5\text{mm}$.

Section 2.0

Cold ISO Alignment

Cold ISO Alignment is done before the tube is heated by Generator calibrations. Hot ISO is done later. For details on Hot ISO, see [Figure 4.0, on page 42](#).

2.1 Verify Tube Temperature < 200° C

For details on how to verify Tube Temperature, see the procedure in section 1.1.

2.2 Accessing the Software

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT
- 3.) Select TUBE CHANGE
- 4.) Select ISO ALIGNMENT

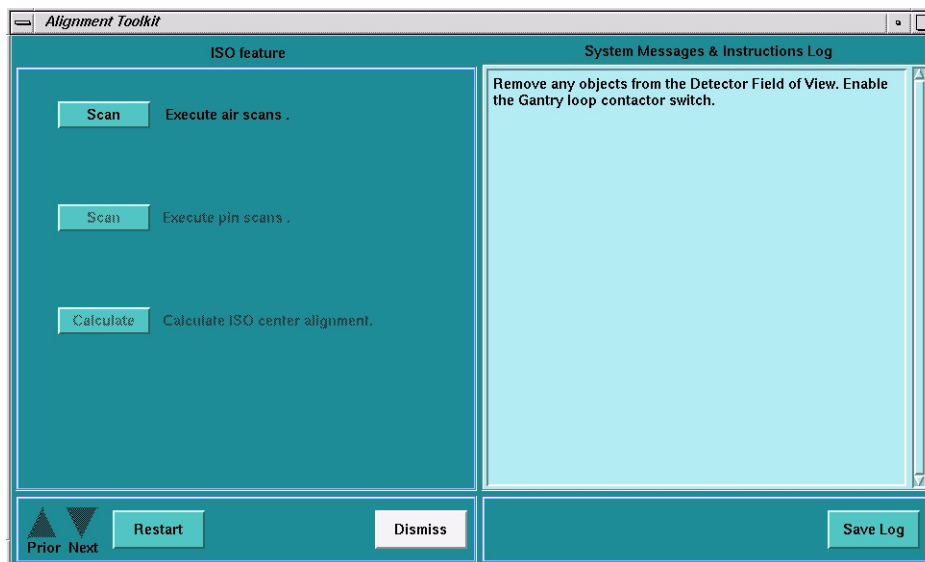


Figure 4-3 ISO Alignment Menu

2.3 ISO Adjustment Procedure

- 1.) Execute Air scan (small spot).
- 2.) Execute Air scan (large spot).
- 3.) Place the 1/8 inch screw driver on the phantom holder (should be pointing into the Z direction).
- 4.) Execute pin scan (small spot).
- 5.) Execute pin scan (large spot).
- 6.) **CALCULATE** ISO center alignment.
- 7.) Mount the dial indicator on the tube assembly. (Figure 4-4.) Make sure that you zero it.



Figure 4-4 ISO Dial Gauge Mounting Location

- 8.) Loosen the 4 M-12 bolts on the tube assembly a half turn.
- 9.) Adjust the tube UP / DOWN as indicated by the calculation. The adjustment bolt is located on the top of the tube - Please see Figure 4-5.

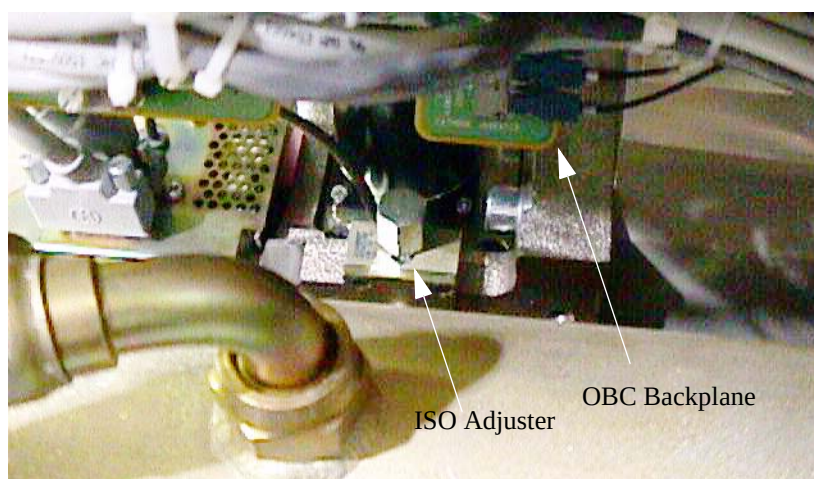


Figure 4-5 ISO Alignment Adjuster

- 10.) Tighten the four (4) M12 bolts and verify dial gauge still reads the correct adjustment value.
- 11.) Repeat steps 1 through 6.
- 12.) If the adjustments are within limit proceed to the next step, otherwise go to step 9.
- 13.) Tighten the four (4) M12 bolts. Torque to 49 lb-ft. (66.4 Nm).
- 14.) If ISO adjustments were made, recheck BOW to verify that the Beam-on-Window values are within the

drift spec of $-1.5\text{mm} \pm 0.5\text{mm}$.

- If BOW is within spec, DO NOT adjust anything
- If BOW is out of spec, Z-align, ISO and BOW recheck will need to be done.

Section 3.0

Calibration - High Voltage

3.1 Access HV Maintenance through Service Desktop

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select HV SUBSYSTEM CHANGE.
- 4.) Select GENERATOR CHARACTERIZATION.

3.2 Generator Characterization

Use the Generator Characterization Program to update the “small spot” and “large spot” characterization files, to provide a starting point for the closed loop mode of the generator. This iterative process requires several scans at a different KV/MA/spot size. It calculates corrections, repeats the scan until the results fall within tolerance, then updates the characterization file.

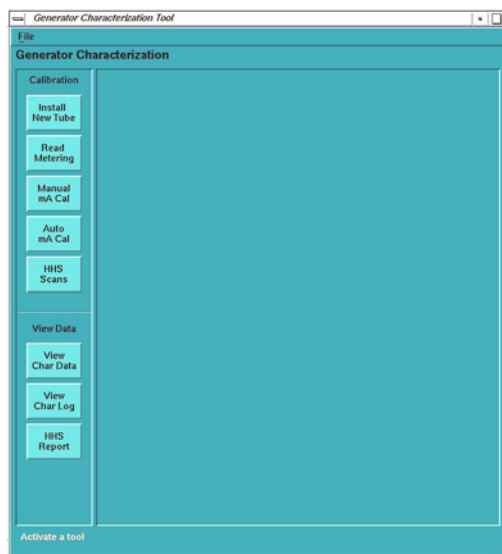


Figure 4-6 Proprietary Generator Characterization Menu Screen

Real Time Information

Patient Handling

Scanning

Delay Timer

Auto mA Calibration

80 kV, 100 mA, 0.1 sec, 0.0mm

0 of 8 Tube calibration stations have completed

Cancel

Pause

Resume

Figure 4-7 Auto mA Calibration Status Screen

3.3 Verify kV Meter

This section describes the calibration check of system internal kV metering circuits.

- 1.) Select READ METERING.
- 2.) Select RUN to start the test. During the test, the firmware reads the metering circuits in the OFF state, then reads the metering circuits in the ON state, and finally reports the readings to the display.
- 3.) Compare the data in the “Delta” column on the Read Meter screen (Figure 9-4) to the data in the “Limit” column.

Note: “Delta” = DVM - A/D

CIRCUIT “OFF”	CIRCUIT “ON”
Anode kV = 0 ± 0.5	Anode kV = 50 ± 7.5
Cathode kV = 0 ± 0.5	Cathode kV = 50 ± 7.5
Total kV = 0 ± 0.5	Total kV = 100 ± 15.0

Table 4-1 Generator Characterization Test Specifications



Figure 4-8 Proprietary Read Metering Screen

3.4 Verify mA Meter – HHS Requirement



WARNING

WHEN SERVICING THE GANTRY:

- NEVER PUT ANY BODY PART INTO THE GANTRY WITHOUT FIRST DISABLING THE AXIAL DRIVE AND RE-VERIFYING (CHECK TWICE) THAT IT IS DISABLED.
- ENSURE THAT THE DRIVE STATUS **LED** (Figure 4-9) IS NOT LIT. DO NOT SERVICE THE GANTRY IF THIS **LED** IS ON.

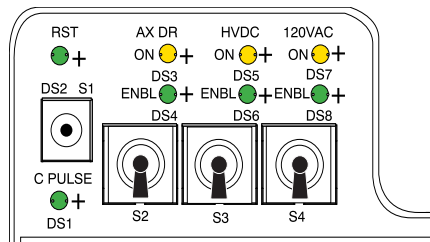


Figure 4-9 STC Backplane Service Switches

This section describes the calibration check system internal mA metering circuits.

- 1.) Inside the Gantry:

- a.) Switch OFF the HVDC ENABLE on **STC** backplane.
- b.) Switch OFF the AXIAL DRIVE ENABLE on **STC** backplane.
- c.) Rotate the Anode tank to the 2 o'clock position.
- d.) Engage the gantry rotational lock.
- 2.) Select SERVICE DESKTOP.
- 3.) Select REPLACEMENT.
- 4.) Select HV SUBSYSTEM CHANGE.
- 5.) Select GENERATOR CHARACTERIZATION.
- 6.) Select READ METERING.

Note: On the display, enter a time delay in seconds, to provide enough time to walk from the console to the DVM, and record the reading. The test will not begin until this time delay expires.
Once it begins, the test enables the meter circuit for only 4 seconds.

- 7.) Use a DVM as an mA meter; connect it to the hardware on the anode side:
 - a.) Connect the black lead to TP8 (ACAL1) on the mA board.
 - b.) Connect the red lead to TP11 (ACAL2) on the mA board.

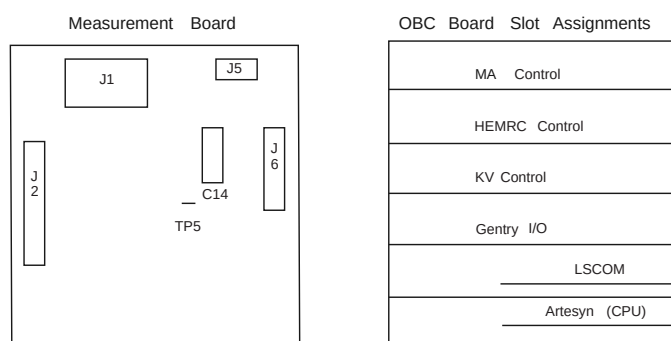


Figure 4-10 Tank Measurement Board

- 8.) On the Display, select the ACCEPT button.
- 9.) Record the displayed, and measured, Anode mA values for "Circuit OFF" and "Circuit ON".

Note: Your system has the test wire to TP5 included in the harness, the Cathode side should read approximately 19 mA during "Circuit On".

- 10.) Disconnect the test equipment from the Anode side, if used.

Note: When you exit Generator Characterization, this test may generate
kV board tube spit counter = x error messages.

- 11.) Use a DVM as an mA meter:
 - a.) Connect the black lead to TP9 (CCAL1) on the mA board.
 - b.) Connect the red lead to TP14 (CCAL2) on the mA board.

- 12.) On the Display, Select the ACCEPT button.

- 13.) Record the displayed, and measured, Cathode mA values for "Circuit OFF" and "Circuit ON".

Note: Your system has the test wire to TP5 included in the harness, the Anode side should read approximately 20mA during "Circuit On".

- 14.) Disconnect the test equipment from the Cathode side if used.

Note: When you exit Generator Characterization, this test may generate
kV board tube spit counter = x error messages.

3.5 KV Gain Pots Adjustment – HHS Requirement

3.5.1 Install HV Divider

- 1.) Inside the Gantry on the **STC** backplane:
 - a.) Switch OFF the **HVDC ENABLE**.
 - b.) Switch OFF the **AXIAL DRIVE ENABLE**.
 - c.) Rotate the Tube to the 3 o'clock position
 - d.) Engage the gantry rotational lock.
 - e.) Switch OFF the **120 VAC GANTRY POWER**.
 - f.) Install the **HV** Divider between Tube and Tanks.

Note: Place the **HV** Divider on a table or tube hoist, so the cables reach the tube.

- 2.) Add a ground wire (minimum size of AWG 12) from Tube ground to bleeder ground. Refer to [Figure 4-11](#).

 **CAUTION** **Potential Electrical Hazard. Performix tube unit MUST be grounded to the gantry during testing.**

- 3.) Switch ON the **120 VAC GANTRY POWER**.
- 4.) Switch ON the **HVDC ENABLE**.

- 5.) Press the  button, on the gantry control panel, turning ON the drives power.

Note: If the gantry covers are removed press the **RESET BUTTON** on the **STC** backplane to turn ON Drives power.

- 6.) Reset the hardware.

 **NOTICE** **Potential for tube damage** **Incorrect installation of anode and cathode HV cables can destroy the Performix tube unit.**

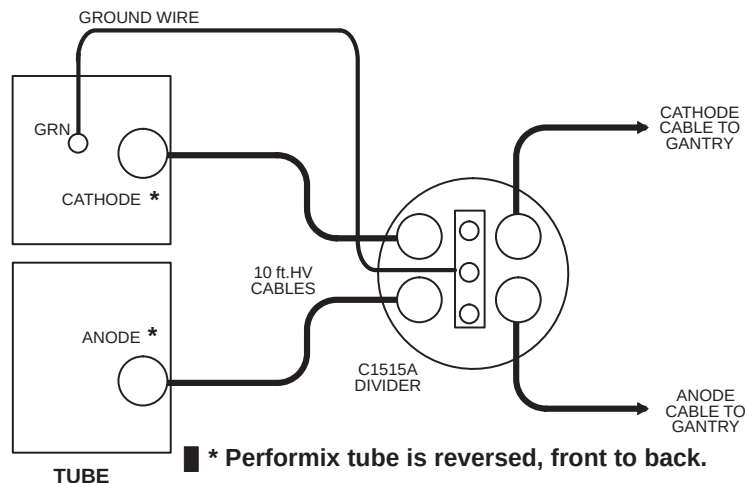


Figure 4-11 HV Divider Installation

3.5.2 Set Up Instrumentation

Use an oscilloscope with 10X probes

- 1.) Use the Gantry Service Outlet to provide 120 Vac power for the scope. This will reduce noise on the scope waveform.
- 2.) Connect **channel one** to the **anode** output of the divider. Connect the scope ground to bleeder ground.

- 3.) Connect **channel two** to the **cathode** output of the divider. Connect the scope ground to bleeder ground.
- Note: In order to minimize bleeder-induced ripple on the kV waveform, connect a 30 foot Belden shielded twisted pair cable between the scope probes and the bleeder.
- 4.) Trigger channel one, positive, DC couple, trigger mode normal.
- 5.) Channel one and two, 10v/div, time base 200ms.
- 6.) Invert Channel two.

3.5.3 Calibrate the Cathode

- 1.) Select SERVICE DESKTOP.
 - 2.) Select REPLACEMENT.
 - 3.) Select HV SUBSYSTEM CHANGE.
 - 4.) Select BLEEDER SETUP (DDC)
 - 5.) Verify/Set-up the following **DDC** parameters:

- <u>STATIC X-RAY ON</u>	Interscan Delay <u>2.00</u>
- <u>1 SECOND</u>	DAS Gain <u>31</u>
- <u>1 SCAN</u>	Gantry Tilt <u>0.0</u>
- <u>FOCAL SPOT LARGE</u>	Trigger Rate <u>984</u>
- <u>100 KV</u>	Filter <u>BLOCKED</u>
- <u>50 MA</u>	Calibration Vector <u>NONE</u>
- <u>MONITOR ENABLE</u>	
 - 6.) Select ACCEPT RX. The Computer Displayed reading specification for the Cathode kV and Anode kV equals 50 ± 0.5 kV.
- Note: If you use scope cursors to window the trace, position the Left Vertical Cursor to the Right of the Rising Edge of the waveform. Position the Right Vertical Cursor to the Left of the Falling Edge of the Waveform.
- 7.) Adjust the Cathode pot on the kV board, until the scope reading for the Cathode kV, and the displayed reading for the Cathode kV in the message log, fall within ± 0.5 kV of each other.
 - 8.) Use the pot, labeled CAKV, R316, on the kV board, to adjust the scope reading.
 - CCW decreases the scope kV.
 - CW increases the scope kV.
 - 1/2 turn equals approximately 0.5 kV.
 - 9.) Record the results on **FORM 4879**.

3.5.4 Calibrate the Anode

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select HV SUBSYSTEM CHANGE.
- 4.) Select BLEEDER SETUP (DDC)
- 5.) Verify/Set-up the following **DDC** parameters:

- <u>STATIC X-RAY ON</u>	Interscan Delay <u>2.00</u>
- <u>1 SECOND</u>	DAS Gain <u>31</u>
- <u>1 SCAN</u>	Gantry Tilt <u>0.0</u>
- <u>FOCAL SPOT LARGE</u>	Trigger Rate <u>984</u>
- <u>100 KV</u>	Filter <u>BLOCKED</u>
- <u>50 MA</u>	Calibration Vector <u>NONE</u>
- <u>MONITOR ENABLE</u>	

- 6.) Select ACCEPT RX. The Computer Displayed reading specification for the Cathode kV and Anode kV is 50 ± 0.5 kV.
- Note: If you use scope cursors to window the trace, position the Left Vertical Cursor to the Right of the Rising Edge of the waveform. Position the Right Vertical Cursor to the Left of the Falling Edge of the Waveform.
- 7.) Adjust the Anode pot on the kV board, until the scope reading for the Anode kV, and the displayed reading for the Anode kV in the message log, fall within ± 0.5 kV of each other.
- 8.) Use the pot, labeled ANKV, R318, on the kV board, to adjust the scope reading.
- CCW decreases the scope kV.
 - CW increases the scope kV
 - 1/2 turn is approximately 0.5 kV.
- 9.) Record the results on **FORM 4879**.

3.5.5 Measure Total kV

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select HV SUBSYSTEM CHANGE.
- 4.) Select BLEEDER SETUP (DDC).
- 5.) Verify/Set-up the following **DDC** parameters:

- <u>STATIC X-RAY ON</u>	Interscan Delay <u>2.00</u>
- <u>1 SECOND</u>	DAS Gain <u>31</u>
- <u>1 SCAN</u>	Gantry Tilt <u>0.0</u>
- <u>FOCAL SPOT LARGE</u>	Trigger Rate <u>984</u>
- <u>100 KV</u>	Filter <u>BLOCKED</u>
- <u>50 MA</u>	Calibration Vector <u>NONE</u>
- <u>MONITOR ENABLE</u>	
- 6.) Change the oscilloscope to add ch.1 and ch.2, to read total kV from the **HV** divider.
- 7.) Channel one and two, 20v/div, time base 200ms, trigger channel. one, positive.
- 8.) Select ACCEPT RX.
- 9.) Record the scope reading, and the Average. kV displayed in the message log, in **FORM 4879**.
- 10.) Display the Generator Characterization menu.
- 11.) Toggle the soft-key MONITOR ENABLE **OFF**, so the message log no longer displays kV and mA readings.

3.5.6 Verify kV Meter

Use this procedure to verify the calibration of the internal kV metering circuits.

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select HV SUBSYSTEM CHANGE.
- 4.) Select GENERATOR CHARACTERIZATION.
- 5.) Select READ METERING.
- 6.) Select ACCEPT.
 - The test begins after the time delay expires.
 - Once the test begins, the software enables the meter circuit for 4 seconds.
- 7.) Record the displayed Anode kV, Cathode kV and Total kV values in the **FORM 4879** "Circuit OFF" and "Circuit ON" table.

- 8.) Select BACKUP.

3.5.7 Remove the External HV Divider

- 1.) Press the  button, on the gantry control panel, turning **OFF** the drives power.

Note: If the gantry covers are removed, proceed to step 2.

- 2.) Inside the Gantry on the **STC** backplane:
 - a.) Switch OFF the HVDC ENABLE.
 - b.) Switch OFF the AXIAL DRIVE ENABLE.
 - c.) Switch OFF the 120 VAC GANTRY POWER.
 - d.) Remove the **HV** Divider between the Tube and Tanks (Figure 4-11, on page 38).
 - e.) Reconnect the **HV** cables for normal operation.



NOTICE

Incorrect installation of anode and cathode HV cables can destroy the Performix tube.

- f.) Re-apply paper toweling around tube locking ring to absorb excess oil.
- g.) Disengage the gantry rotational lock.
- a.) Switch ON the HVDC ENABLE.
- b.) Switch ON the AXIAL DRIVE ENABLE.
- c.) Switch ON the 120 VAC GANTRY POWER.

- 3.) Press the  button, on the gantry control panel, turning ON the drives power.

Note: If the gantry covers are removed, press the RESET BUTTON on the **STC** backplane to turn ON Drives power.

- 4.) Reset the hardware.

3.5.8 Install New Tube Program – HHS Requirement

Use this program to complete Auto mA Cal on a new tube. *Run this program only on a new tube.*

Refer to Figure 9-5.

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select HV SUBSYSTEM CHANGE.
- 4.) Select GENERATOR CHARACTERIZATION.
- 5.) Select the INSTALL NEW TUBE.

Note: The system automatically warms up the tube.

- 6.) The system prompts you with the tube type. Verify that the number corresponds to your tube type; answer **Y** or **N**. The tube should show as 15-MX_200CT. If it does not, contact the Online Center.
- 7.) Press START SCAN when it flashes, to automatically run the program and update the display:
seed filament current shift scans

3.5.9 Auto mA Calibration

Run this program when you replace the X-Ray tube, or the system requires re-calibration.

Note: If you have just completed running the Install New Tube Program in the previous section, you do NOT need to run Auto mA Cal.

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.

- 3.) Select HV SUBSYSTEM CHANGE.
- 4.) Select GENERATOR CHARACTERIZATION.
- 5.) Select AUTO MA CAL.

Note: The software automatically warms up the tube.

- 6.) Press START SCAN when it flashes, to automatically run the program and update the display:
 - Ductility warm-up -
 - Auto mA Cal -
- 7.) The system displays the final filament currents on the screen.

3.5.10 kV AND mA Verification Data Sheet – HHS Requirement

Access the software and perform the following:

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select TUBE CHANGE.
- 4.) Select HHS SCANS.

When Scans are complete:

- 5.) Select HHS REPORT.

Insure all kV and mA stations meet specifications found in form 4879. (Form can be found in the appropriate system Advanced Service manual.)

Note: If your system has InSite connectivity, you can have a copy of the HHS report emailed to your laptop for printing by sending a sweep report to the autoSC.

Send an email in the following format:

TO: autosc (MED) OR autosc@olc.med.ge.com

SUBJECT: sweep

Message content:

type = get_hhs_report

system id = xxxxx

Section 4.0 HOT ISO Alignment

4.1 Accessing the Software

- 1.) Select SERVICE DESKTOP.
- 2.) Select CALIBRATION.
- 3.) Select HOT ISO ALIGNMENT.

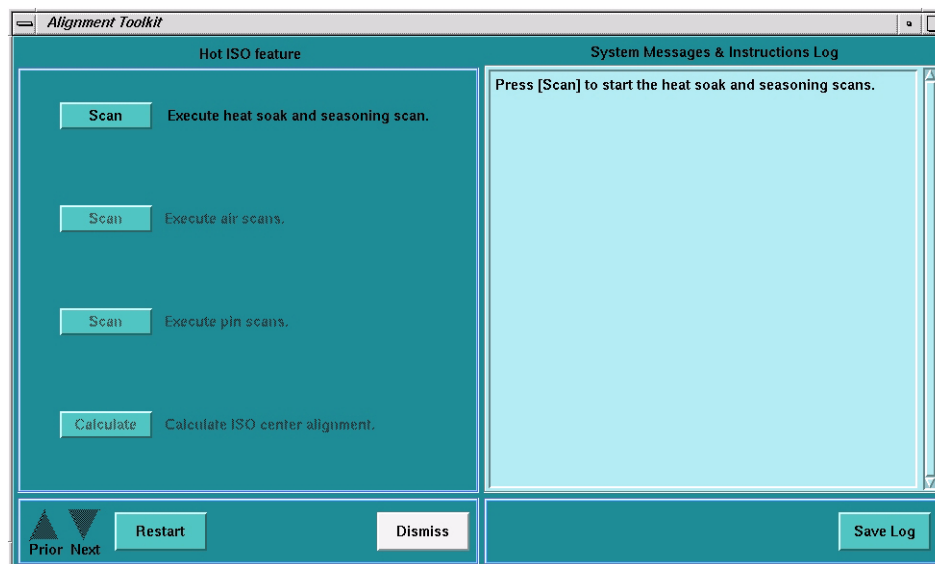


Figure 4-12 HOT ISO Screen

4.2 Adjustment Procedure

- 1.) Execute the Heat Soak and Seasoning scan.
- 2.) Execute Air Scan (small spot).
- 3.) Execute Air Scan (large spot).
- 4.) Place the 1/8 inch screw driver on the phantom holder (should be pointing into the Z direction).
- 5.) Execute Pin Scan (small spot).
- 6.) Execute Pin Scan (Large spot)
- 7.) Calculate correction factor with Calculate button.

Note: No movement of the tube is required. This is a software correction done automatically.

Section 5.0 New Tube Configure

Access the software and perform the following:

- 1.) Select SERVICE DESKTOP.
- 2.) Select REPLACEMENT.
- 3.) Select TUBE CHANGE.
- 4.) Select NEW TUBE CONFIG.
- 5.) Select SCAN HARDWARE RESET and reset application SCAN.

Section 6.0 Collimator Calibration

- 1.) Perform Collimator Calibration by selecting the function from Scanner Utilities (**left head**).
- 2.) Select SCANNER UTILITIES, and Select COLLIMATOR CALIBRATION.

- 3.) Ensure that there is nothing in the x-ray beam and continue.
- 4.) The system will now perform a Mylar window check if needed and provide the appropriate messages if the window should need cleaning.
- 5.) Upon completion of the Mylar window scans the system will now take 8 scans and save the results in the Calibration Data Base. The system will provide the appropriate messages if the calibration should fail.

Section 7.0

Detector Z-Slope Calibration

- 1.) Perform Z-Slope Calibration by selecting the function from Scanner Utilities.
- 2.) Select SCANNER UTILITIES, and select Z-SLOPE CALIBRATION.
- 3.) Ensure there is nothing in the x-ray beam and continue.
- 4.) The system will now scan and provide the appropriate messages if the calibration should fail.

Section 8.0

DAS Gain Calibration

- 1.) Perform **DAS** Gain Calibration by selecting the function from Scanner Utilities (**left head**).
- 2.) Select SCANNER UTILITIES, and Select DAS GAIN CALIBRATION.
- 3.) Ensure that there is nothing in the x-ray beam and continue.
- 4.) The system will now perform a Mylar window check and provide the appropriate messages if the window should need cleaning.
- 5.) Upon completion of the Mylar window scans the system will now take 31 scans and save the results in the Calibration Data Base. The system will provide the appropriate messages if the calibration should fail.

Section 9.0

Calibration Process



CAUTION
Potential for
IQ Artifacts

If kV Gain Pots were adjusted in section 3.5, you cannot use Fastcal. A full set of calibrations must be performed.

- 1.) Perform a Gantry Balance check.
 - a.) SERVICE DESKTOP > REPLACEMENT > GANTRY SERVICE BALANCE
 - b.) Check that the tube counterbalance weights are installed and bolts are secure.
 - c.) Complete the Gantry balance check.
- 2.) If this test does not pass, it is not recommended that you perform calibrations until the balance issue is resolved. Refer to the balance procedure in the Gantry Section of the appropriate Advanced Service Manual.
- 3.) Bring up the main menu: It is represented as an icon located on the bottom of the screen labeled as “Daily Prep”. Click the on-screen DAILY PREP button (**left head**).



CAUTION
Potential for
IQ Artifacts

If Fastcal has been run in the past 48 hours, only one filament will be selected. In this case, Fastcal will need to be run **twice** in order to calibrate both filament sections. It is suggested that you watch the Fastcal scanlist as scans are being taken to insure that both filaments are being scanned.

Note:

To check when the last Fastcal was run:

- a.) Open a unix shell.
- b.) `cd /usr/g/service/state`
- c.) `tail -50 ssw.calreport.hist`
- 4.) Select and run FASTCALIBRATION.
- 5.) The system will run through the Normal Fast Calibration Procedure.

Section 10.0

CT N# Adjust

- 1.) After the Fast Calibration is run, run CT N# Adjust Scans.
- 2.) Select SCANNER UTILITIES, and select AUTO CT NUMBER ADJUST.
- 3.) Place the 20 cm water phantom on the phantom holder and center the phantom.
- 4.) Run the CT N# Adjust scans.



CAUTION
Potential for IQ
artifacts

Because DETAILED CALIBRATIONS were not run, it is IMPERATIVE to visually inspect all CT N# Adjust Images for artifacts.

- 5.) After all Scans have completed, visually inspect all images for artifacts. If artifacts are seen, take note of the scan configuration, i.e. 4 x 2.5 / 200 mA / 120 kV / 2s

Section 11.0

Quick Tube Change IQ Scans and Image Analysis

11.1 Quick Tube Change IQ Scans – Large FOV

Purpose: Because Auto CT Number Adjust scans only look at the small field of view, there needs to be a series of scans taken with the large field of view. The 35cm and 48cm Phantoms will be used. Here is a listing of scans, and the procedure to run Image Analysis.

TEST	PROTOCOL	SERIES DESCRIPTION	4-SLICE	8-SLICE	16-SLICE
QA3 - 120kV - 10mm - 2i	Image Series QA	QA#3/Small/1.0s	X		
QA3 - 120kV - 8x2.50		QA#3/Small/8x2.5		X	
QA3 - 120kV - 16x0.625		QA#3/Small/16x0.625			X
QA3 - thintwin or 0.625/2i		QA#3/Small/2x0.625/2s	X	X	X
P35 - 4x1.25	Image Series 35cm Poly	35cm/4x1.25/2s	X		
P35 - 8x1.25		35cm/8x1.25/2s		X	
P35 - 16x1.25		35cm/16x1.25/2s			X
P48 - 4x2.5	Image Series 48cm Poly	48cm/4x2.5mm/1s	X		
P48 - 8x2.5/120/200/1s		48cm/8x2.5mm/1s		X	X

Table 4-2 Quick Tube Change IQ Scans

- 1.) The phantom under test must be centered before starting scanning.
- 2.) Select SCANNER UTILITIES and CENTER PHANTOM.
- 3.) Repeat using the Center Phantom function until the phantom is centered.
- 4.) Exit the Center Phantom Utility and The Scanner Utilities Screen.
- 5.) Select NEW PATIENT.
- 6.) Enter a PATIENT ID, and PATIENT NAME.
- 7.) Select the SERVICE tab from the Anatomical Selector.
- 8.) Select MANUFACTURING.
- 9.) Select the Protocol being investigated (Image Series QA, Image Series 35 cm Poly or Image Series 48 cm).
- 10.) Hit the NEXT SERIES button until the desired system configuration appears.
- 11.) Scan the series and move on to the next desired system configuration.
- 12.) When the IQ scans are done for a phantom, select END EXAM.
- 13.) Return to Step 1 and repeat for all phantoms being scanned.
- 14.) When all Phantoms have been scanned from the Service Desktop, select IMAGE QUALITY and IMAGE ANALYSIS.
- 15.) Select the small AUTO 1X button at the top of the screen.
- 16.) Select the pull down menu AUTO 1X, select the Phantom to Analyze (ImgSer 48cm or ImgSer 35cm), and select the Scan Configuration to Analyze, and select the AUTO button.
- 17.) In the SORT WINDOW, make sure the exam being Analyzed is highlighted.
- 18.) Back in the Image Analysis Window click on the ACCEPT button.
- 19.) A PASS/FAIL window will appear, along with details of what Passed and Failed.
- 20.) If the scan passed, that system configuration is in an acceptable state. Repeat steps 16 through 19 until all scans have been analyzed.

- 21.) If any scans fail, take note of the system configuration.

11.2 Image Quality Check for Form 4879 - HHS Requirement

In addition to the Quick Tube Change IQ Scans, you will also need to refer to the FORM 4879 that is appropriate for your scanner and scan the phantoms required at the prescribed techniques in order to complete FORM 4879.

Section 12.0 Artifacts and Image Analysis Failures

Purpose: If there were no Artifacts or Image Analysis Failures, the Quick Tube Replacement Procedure is complete. Follow the following instructions if you have Artifacts or Failures.

- 1.) Compile your notes from CT N Adjust Artifacts and Image Analysis Failures.
- 2.) Select SCANNER UTILITIES and DETAILED CALIBRATION.
- 3.) A Gantry Balance Check will run.
- 4.) From the Detailed Calibration list, select only the scan configurations that failed.

Example: There was a CT Number Adjust Artifact at 120kV with a 4x1.25 scan configuration, and there was an Image Analysis Failure at 120kV at 4x2.5, the 120KV button and the 4x1.25 and 4x2.5 slice collimation buttons would be selected. Leave all of the SFOV and FOCAL SPOT buttons selected.

- 5.) Run the Detailed Calibrations.
- 6.) Once Detailed Calibrations are done, re-run AUTO CT NUMBER ADJUST. Run the same system configurations that were run for Detailed Calibrations.
- 7.) Check to ensure that the CT Number Adjust scans are free of artifacts and make sure that Image Analysis now passes on the previously failed configurations.

Section 13.0

Gantry Rotation Safety Check



WARNING

This test proves that the scanner is safe to rotate at maximum rotation speed.

CRUSH HAZARD.

EQUIPMENT COULD EJECT FROM GANTRY.

ENSURE THAT NO PERSONNEL ARE IN THE SAME ROOM/AREA AS THE GANTRY, PRIOR TO AND DURING THIS SCAN.

Use DDC to Rotate the Gantry for 10 minutes, at the fastest allowed gantry speed.

- 1.) Launch Diagnostic Data Collection (DDC) Utility:
 - a.) Open the Common Service Desktop
 - b.) Click on the DIAGNOSTICS Icon
 - c.) Click on the DATA ACQUISITION Folder
 - d.) Click on the DIAGNOSTIC DATA COLLECTION Menu Item
- 2.) Prepare System for DDC Scan. Refer to [Figure 4-13](#).
 - a.) Select ROTATING X-RAY OFF
 - b.) Type: **120** in Scan Time (sec) field
 - c.) Type: **0** in the Gantry Velocity (sec/rev) field and press ENTER. The system will then default to the fastest allowable service mode revolution time.
 - d.) No of Scans = **5**. This will result in 10 minutes of rotation.
 - e.) Verify other settings, as shown.
 - f.) Click on ACCEPT RX button, to send scan request to firmware

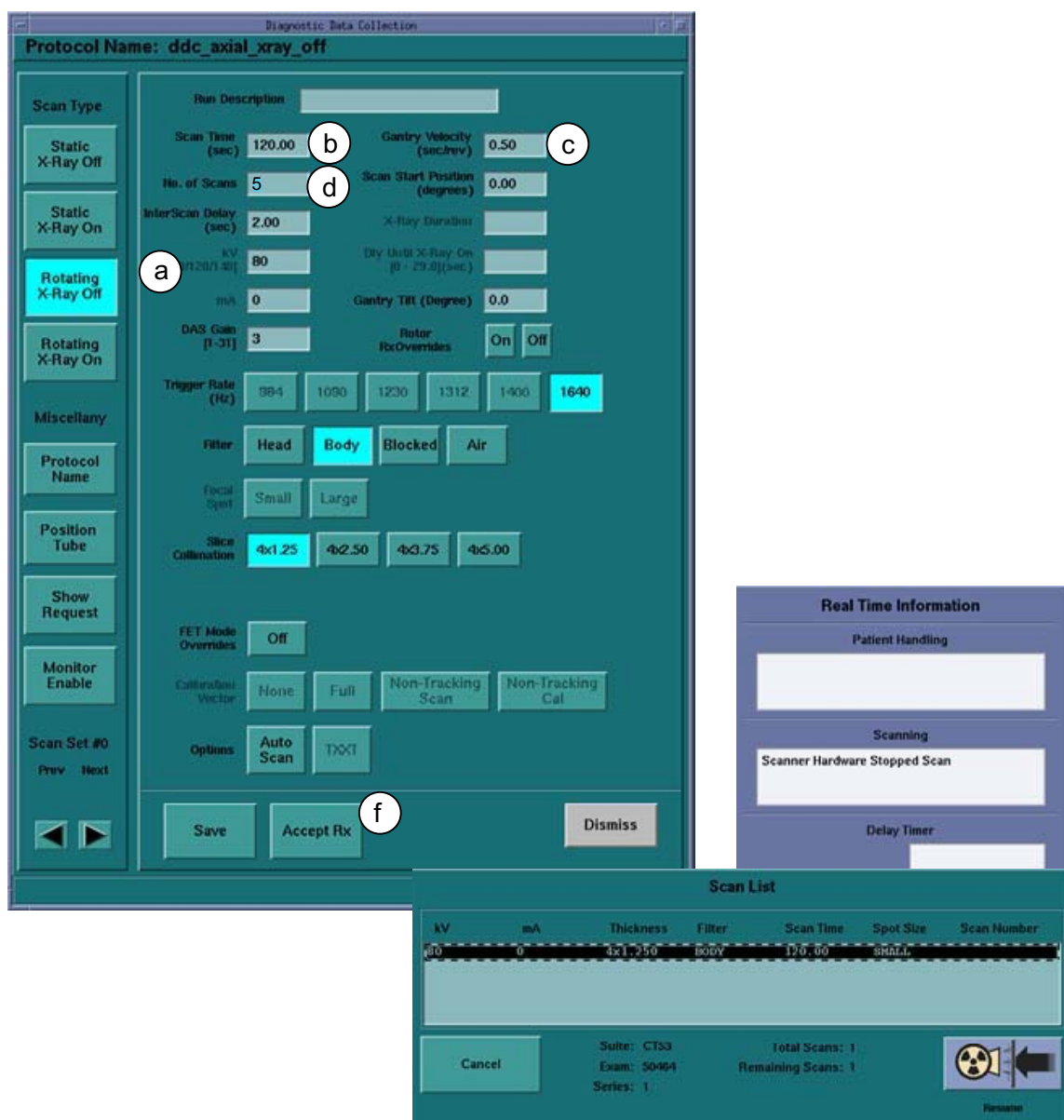


Figure 4-13 DDC Tool

The Utility:

- Displays the Scan List GUI and the Real Time Information window.
 - After a few seconds, it starts flashing the SCAN button on the console.
- 3.) Press the Scan button on the Console, to start the DDC Scan. When complete, the DDC Utility dismisses the Scan List GUI and the Real Time Information window.
 - 4.) If no issues arise, gantry rotation capability is safe for use.

Section 14.0 Finalization

Save system state.

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